

Mark schemes

Q1.

(a) Volume $X \sim N(56, 2.5^2)$

$$P(X < 60) = P\left(Z < \frac{60 - 56}{2.5}\right)$$

Standardising (59.5, 60 or 60.5) with 56

and ($\sqrt{2.5}$, 2.5 or 2.5^2) and/or $(56 - x)$

M1

(i) $= P(Z < 1.6)$

CAO; ignore sign

A1

$= 0.945$

AWRT (0.94520)

A1

3

(ii) $P(50 < X < 60) = (i) - P(X < 50)$

Or equivalent

M1

$= (i) - P(Z < -2.4) = (i) - [1 - \Phi(2.4)]$

Area change

m1

$= 0.94520 - (1 - 0.99180) = 0.937$

AWRT (0.93700)

A1

3

(iii) $P(X = 55) = 0$

CAO

B1

1

(b) $98\% \Rightarrow z = 2.05 \text{ to } 2.06$

AWFW; ignore sign (-2.0537)

B1

$$z = \frac{100 - \mu}{3.4}$$

Standardising 100 with μ & 3.4
Allow $(\mu - 100)$

M1

Thus $\frac{100 - \mu}{3.4} = -2.0537$

Equating z-term to z-value
Not using 0.98, 0.02 or $|1 - z|$

M1

Thus $\mu = 107$
 AWR

A1

4

[11]

Q2.

Weight, $X \sim N(205, 25^2)$

(a) (i) $P(X < 250) = P\left(Z < \frac{250 - 205}{25}\right)$

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Standardising (249.5, 250 or 250.5)
with 205 and ($\sqrt{25}$, 25 or 25^2) and/or $(205 - x)$

M1

$= P(Z < 1.8)$
CAO; ignore sign

A1

$= 0.964$
AWRT (0.96407)

A1

3

(ii) $P(200 < X < 250) = (i) - P(X < 200)$
Or equivalent

M1

$$\begin{aligned}
 &= (i) - P(Z < -0.2) = (i) - [1 - \Phi(0.2)] \\
 &\quad \text{Area change} \\
 &= 0.96407 - (1 - 0.57926) = 0.543 \\
 &\quad \text{AWRT (0.54333)}
 \end{aligned}$$

A1

3

(b) (i) $(100 - 30)\% = 70\% \Rightarrow z = 0.524 \text{ to } 0.525$
 AFWW; ignore sign (-0.5244)

B1

$$\text{Thus } \frac{s - 205}{25} = -0.5244$$

*Equating z-term, involving 205 and 25,
 to z value
 Not using 0.3, 0.7 or $|1 - z|$
 Allow $(205 - s)$*

M1

$$\text{Thus } s = 191.9$$

AWRT

A1

(ii) $(100 - 20)\% = 80\% \Rightarrow z = 0.841 \text{ to } 0.842$
 AFWW, ignore sign (0.8416)

B1

$$\text{Thus } \frac{m - 205}{25} = 0.8416$$

*Only if not awarded in (i)
 Not using 0.2 or 0.8
 Allow $(205 - m)$*

(M1)

$$m = 226.0$$

AWRT; accept 226

A1

5

(c) $90\% \Rightarrow z = 1.28$

AWRT; ignore sign (1.2816)

B1

$$z = \frac{200 - 175}{\sigma}$$

*Standardising 200 with 175 & σ
Do not allow $175 - 200$*

M1

Thus $\frac{25}{\sigma} = 1.2816$

*Equating z-term to z-value
Not using 0.9 or 0.1*

m1

Thus $\sigma = 19.5$

AWRT

A1

4

[15]

Q3.

(a) Length $X \sim N(48, 8^2)$

$P(X = 60) = 0$

CAO

B1

1

(b) $P(X = 60) = P\left(Z < \frac{60 - 48}{8}\right)$

standardising (59.5, 60 or 60.5) with 48

and $(\sqrt{8}, 8 \text{ or } 8^2)$ and/or $(48 - x)$

M1

$P = (Z < 1.5) = 0.933$

AWRT (0.93319)

A1

2

(c) $5\% \text{ of } 48 = 2.4$

CAO; OE

B1

$$P(48 - x < X < 48 + x)$$

*attempt at the probability of an interval,
symmetric about 48*

M1

$$= P(-0.3 < Z < 0.3)$$

CAO 0.3

A1

$$= \Phi(0.3) - (1 - \Phi(0.3))$$

area change

m1

$$= 2 \times 0.61791 - 1$$

$$= 0.235 \text{ to } 0.236$$

AWFW (0.23582)

A1

5

[8]

Q4.

(a) $L \sim N(10.25, \sigma^2)$

$$P(L < 10) = P\left(Z < \frac{10 - 10.25}{\sqrt{0.04}}\right) =$$

*standardising (9.5, 10 or 10.5) with
($\sqrt{0.04}$, 0.04 or 0.04^2) and/or $(10.25 - 10)$*

M1

$$P(Z < 1.25) =$$

cao; ignore sign

A1

$$1 - \Phi(1.25) =$$

area achange

m1

$$1 - 0.89435 =$$

0.105 to 0.106

awfw

A1

4

(b) $P(L > 10) = 0.98$

$z_{0.98} = -2.0537$

awfw 2.05 to 2.06; ignore sign

B1

Also $z = \frac{10 - 10.25}{\sigma}$

*standardising (10 or 10.5) with 10.25 and σ ;
allow (10.25 – 10)*

M1

Thus $\frac{10 - 10.25}{\sigma} = -2.0537$

equating z-term to z-value; not using 0.98, 0.02 or $|1 - z|$

m1

Thus $\sigma = 0.121$ to 0.122

*awfw; do **not** ignore sign
(A0 if negative sign dropped)*

A1

4

[8]

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Q5.

(a) (i) Weights $W \sim N(25.8, 0.5^2)$

$$P(W < 25) = P\left(Z < \frac{25.0 - 25.8}{0.5}\right)$$

M1

$$= P(Z < -1.6) = P(Z > 1.6) = 1 - \Phi(1.6)$$

Area change

m1

$$1 - 0.94520 = 0.05480 \Rightarrow 0.054 \text{ to } 0.055$$

AWFW

A1

(ii) $P(25.5 < W < 26.5)$

$$P\left(\frac{25.5 - 25.8}{0.5} < Z < \frac{26.5 - 25.8}{0.5}\right) = P(-0.6 < Z < 1.4)$$

CAO both; ignore signs

A1

$$\Phi(1.4) - (1 - \Phi(0.6)) = 0.91924 - 1 + 0.72575 = 0.64499$$

m1

$$\Rightarrow 0.644 \text{ to } 0.646$$

AWFW

A1

(b) $P(W > w) = 75\%$

$$z_{0.75} = -0.6745$$

AWFW ± 0.674 to ± 0.675 not 0.67 or 0.68

B1

$$\text{and } z = \frac{w - 25.8}{0.5}$$

Standardising w with $\sqrt{0.5}$, 0.5 , or 0.5^2 and/or $(25.8 - w)$

M1

$$\therefore \frac{w - 25.8}{0.5} = -0.6745$$

Equating z-value & z-term; not using 0.75 , 0.25 or $1-z$

m1

$$\therefore w = 25.46 \text{ to } 25.47$$

AWFW; can be gained from using -0.67 or -0.68

A1

Q6.

(a) Time $X \sim N(134, 16^2)$

(i) $P(X < 150) = P\left(Z < \frac{150 - 134}{16}\right)$

standardising (149.5 150 or 150.5 with $\sqrt{16}$, 16 or 16^2 and/or $(134 - x)$

M1

$= P(Z < 1)$
CAO; ignore sign

A1

$= 0.841(34)$
AWRT

A1

3

(ii) $P(X > x) = 10\%$
 $z_{0.10} = 1.2816$

AWFW 1.28 to 1.29; ignore sign

B1

and $z = \frac{x - 134}{16}$

standardising x with $\sqrt{16}$, 16 or 16^2 and / or $(134 - x)$

M1

$\therefore \frac{x - 134}{16} = 1.2816$

equating z-value to z-term not using 0.10, 0.90 or $|1 - z|$

m1

$\therefore x = 154.5$
AWRT

A1

4

(b) Time $Y \sim N(\mu, \sigma^2)$
 $z_{0.14} = -1.0803$

AWRT -1.08, ignore sign; not -1.085

B1

$z_{0.03} = +1.8808$

AWRT + 1.88; ignore sign; not + 1.885

B1

$$P(Y < 170) = P\left(Z < \frac{170 - \mu}{\sigma}\right)$$

and

$$P(Y > 200) = P\left(Z > \frac{200 - \mu}{\sigma}\right)$$

use of either; not with ± 0.5

M1

$$\therefore 170 - \mu = -1.0803\sigma$$

$$\text{and } 200 - \mu = +1.8808\sigma$$

simultaneous equations involving μ , σ and z-values; not probabilities etc

m1

$$\therefore 30 = 2.9611\sigma$$

$$\therefore \sigma = 9.9 \text{ to } 10.2$$

AWFW

A1

$$\therefore \mu = 180.8 \text{ to } 181.2$$

AWFW

A1

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Q7.

$$L \sim N(208.5, 2.5^2)$$

$$(a) \quad P(205 < L < 210)$$

$$= P\left(\frac{205 - 208.5}{2.5} < Z < \frac{210 - 208.5}{2.5}\right)$$

*Standardising (204.5, 205, 205.5 or 209.5, 210, 210.5)
with f.t. 2.5, 2.5 or 2.5²; allow (208.5 - l)*

M1

$$= P(-1.4 < Z < 0.6)$$

CAO; either ignoring sign

A1

$$= \Phi(0.6) - (1 - \Phi(1.4))$$

Area difference

m1

$$= 0.72575 - (1 - 0.91924)$$

$$= 0.72575 - 0.8076 = 0.64499 \text{ or } 0.6448 \text{ to } 0.6452$$

AWFW; AG of 0.645

A1

4

(b) Binomial

Use of

M1

$$P(R = 6)$$

$$= \binom{10}{6} (0.645)^6 = ((1 - 0.645) = 0.355)^4$$

Correct expression; may be implied

A1

$$= 210 \times 0.0720 \times 0.01588 = 0.239 \text{ to } 0.241$$

AWFW

A1

3

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[7]

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Q8.

(a) (i) $A \sim N(1212, 5^2)$

$$P(A > 1205) = P\left(Z > \frac{1205 - 1212}{5}\right) =$$

Standardising 1204.5, 1205, 1205.5 with
 $\sigma, \sigma^2, \sqrt{\sigma}$ and/or $(\mu - a)$

M1

$$P(Z > -1.4) = P(Z < 1.4) =$$

CAO ± 1.4
Area change

A1/m1

0.92

AWRT

A1

4

(ii) $P(X > x) = \frac{1}{500}$

Standardising x with $\sigma, \sigma^2, \sqrt{\sigma}$ and/or $(\mu - x)$

M1

$$\therefore P\left(Z > \frac{x - 1212}{5}\right) =$$

0.002

CAO

B1

z - value = 2.8782

AWRT 2.88

- 2.88 B0

B1

$$\therefore \frac{x - 1212}{5} = 2.8782$$

Equating standardised x - value to z - value

m1

$$\therefore x = 1226$$

AWRT

ignore 1230 if rounded from 1226

1230 without 1226 AO

A1

5

(b) $P(Y < 1200) = 1.5\%$

Attempted use of with attempt at finding
a z -value

M1

$$P(Y > 1225) = 1.9\%$$

Standardising on μ and σ
(must be 1200 and 1225)

m1

$$\therefore \frac{1200 - \mu}{\sigma} = -2.1701$$

AWRT ± 2.17

B1

$$\therefore \frac{1225 - \mu}{\sigma} = +2.0749$$

AWRT ± 2.07 to ± 2.08

B1

$$\therefore 25 = 4.245\sigma$$

Attempted solution of two equations (reduced to one variable)

M1

$$\therefore \sigma = 5.9$$

AWRT (not 6)

A1

$$\therefore \mu = 1212.8$$

AWRT (not 1213)

A1

7

[16]

Q9.

$$(a) \quad (i) \quad z = \frac{11 - 7.5}{2.5} = 1.4$$

method for z – ignore sign

M1

a correct use of normal tables

M1

$$P(<11) = 0.919$$

0.919 (0.919 ~ 0.92)

A1

3

$$(ii) \quad z_1 = \frac{5.5 - 7.5}{2.5} = -0.8$$

method for z's – both signs correct

M1

correct methods, their z's

M1

$$z_2 = \frac{10.5 - 7.5}{2.5} = 1.2$$

Probability between 5.5 and 10.5 is
 $0.88493 - (1 - 0.78814) = 0.673$
completely correct method

m1

0.673 (0.6725 ~ 0.6735) implies full marks

A1

4

(b) $7.5 - 1.2816 \times 2.5 = 4.30$
(1.28 – 1.29)

B1

(their z) × 2.5

M1

completely correct method

m1

4.30 (4.29 ~ 4.3)

A1

4

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[11]

Q10.

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(a) (i) $z = \frac{500 - 506}{5} = -1.2$
method for z – ignore sign

M1

probability < 500 = $1 - 0.88493$
any correct use of normal tables -
generous

M1

= 0.115
0.115 (0.1145 ~ 0.1155)

A1

$$(ii) \quad z_1 = \frac{495 - 506}{5} = -2.2$$

$$z_2 = \frac{505 - 506}{5} = -0.2$$

method both z's ignore sign

M1

both signs correct

m1

probability between 495 and 505
 = 0.98610 – 0.57926
completely correct method

M1

= 0.407
 0.407 (0.406 ~ 0.4075)

A1

4

$$(b) \quad 506 - 3.0902 \times 5 = 490.5g$$

3.0902 or 3.09

B1

(their z) × 5

M1

completely correct method

m1

490.5 (490 ~ 491)

A1

4

[11]

Q11.

$$(a) \quad z = \frac{20 - 26}{8} = -0.75$$

method for z – ignore sign

M1

probability no need to refill

completely correct method

M1

$$1 - 0.77337 = 0.227$$

0.227 (0.226 to 0.227)

A1

$$z = \frac{40 - 26}{8} = 1.75$$

probability exactly one refill i.e. between

M1

$$20 \text{ and } 40 = 0.95994 - 0.2263 = 0.733$$

correct method
0.733 (0.733 to 0.734)

A1

5

(b) $0.22623^5 = 0.000598$

0.000598 (0.00059 to 0.00061)

M1

A1

2

(c) $\mu - 0.9945\sigma = 20$

0.9945 (0.994 to 0.995)

B1

$$\mu - 1.175\sigma = 40$$

1.175 (1.17 to 1.18)

B1

Good attempt at equations - ignore sign

M1

Completely correct equations

m1

$$2.169\sigma$$

Method of solution

m1

$$\sigma = 9.222$$

A1

6

$$\mu = 29.2$$

$$\sigma = 9.222 \text{ (9.21 to 9.23)}$$

$$\text{and } \mu = 29.2 \text{ (29.1 to 29.3)}$$

[13]

Q12.

$$(a) \quad z = \frac{172.5 - 168}{4.5} = 1.0$$

method of standardising – ignore sign

M1

$$P(<172.5) = 0.841$$

A correct use of normal tables

$$0.841 \text{ (0.841, 0.842)}$$

M1 A1

3

$$(b) \quad z_1 = \frac{159 - 168}{4.5} = -2.0$$

method of standardising – consistent signs

M1

$$z_2 = \frac{163.5 - 168}{4.5} = -1.0$$

Signs of z clearly correct

m1

Probability between 159 and 163.5 is

M1

$$0.97725 - 0.84134 = 0.136$$

Correct method – depends on M1 only

$$(0.1355, 0.1365)$$

A1

4

[7]

Q13.

(a) (i) $z_1 = \frac{475 - 490}{10} = -1.5$
Method for z - ignore sign

M1

$z_2 = \frac{510 - 490}{10} = 2.0$
Both z's correct sign

m1

Probability between 475 and 510 =
 $0.97725 - (1 - 0.93319) = 0.910$
A correct use of normal tables - generous
Completely correct method
0.910 (0.91 - 0.911)

M1 m1 A1

5

(ii) $490 - 1.8808 \times 10 = 471.2$ m1
1.8808 (1.88 - 1.9)
Candidate's z $\times 10$
Completely correct method
471.2 (471.1 - 471.2 or 471)

B1 M1 m1 A1

4

(b) $\mu - 1.0364 \times 10 = 475$
1.0364 (1.03 - 1.04)
Candidate's z $\times 10$

B1 M1

$\mu = 485.4$
Completely correct method
485.4 (485.3 - 485.4 or 485)

m1 A1

4

[13]

Q14.

(a) $z_1 = \frac{18 - 19.5}{1.2} = -1.25$

$$z_2 = \frac{16.5 - 19.5}{1.2} = -2.5$$

Method for z ignore sign

M1

Both z's correct

A1

$$p(1.65 < X < 18) = 0.99379 - 0.89435$$

A correct use of normal tables

M1

Completely correct method

m1

$$= 0.0994$$

0.0994 (0.099 to 0.1)

A1

5

(b) $\mu + 0.8416\sigma = 18$

0.8416 and 1.7507

B1

$$\mu + 1.7507\sigma = 19$$

Their $z \times \sigma$

B1

$$0.9091\sigma = 1$$

Both correct equations ignore signs All rights reserved.

B1

$$\sigma = 1.10 \text{ min}$$

Method for σ

M1

$$\mu = 17.1 \text{ min}$$

Method for μ

m1

1.10 (1.09 to 1.11) and 17.1 (17 to 17.1)

A1

6

Q15.

(a) (i) $z = \frac{75 - 85}{8} = -1.25$

Method for z; ignore sign

M1

$P(< 75) = 1 - 0.89435$

A correct use of normal tables

M1

$= 0.106$

0.106 (0.105 to 0.106)

A1

3

(ii) $z_2 = \frac{81 - 85}{8} = -0.5$

Completely correct method; allow both z's positive

M1

Probability between 75 and 85 is

$0.89435 - 0.69146 = 0.203$

Reasonable attempt, both z's negative

M1

0.203 (0.202 to 0.204)

A1

3

(b) $85 + 3.0902 \times 8 = 110$

3.09 or 3.0902

B1

(their z) $\times$ 8

M1

Completely correct method

m1

110 (109 to 110)

A1

4

(c)
$$z = \frac{81 - 85}{\frac{8}{\sqrt{4}}} = -1$$

Use of $\frac{8}{\sqrt{4}}$

M1

Correct method for z

m1

Probability mean less than 81 = $1 - 0.84134 = 0.159$

Completely correct method

m1

0.159 (0.158 to 0.16)

A1

4

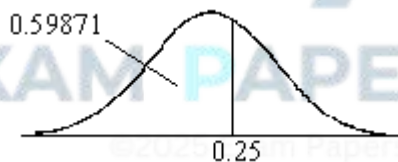
[16]

Q16.

(a) (i)
$$z = \frac{(30 - 28)}{8} = 0.25$$

Method for z - ignore sign

M1



Any correct use of normal tables

M1

Completely correct method

m1

4

(ii) $P(>30 \text{ mins}) = 0.599$

Method for z's - ignore signs if consistent

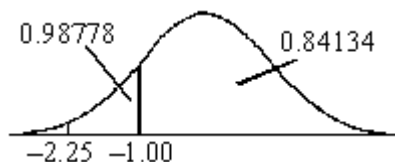
M1

$$z = \frac{10 - 28}{8} - \frac{20 - 28}{8} = -2.25$$

$$z = \frac{10 - 28}{8} = \frac{20 - 28}{8} = -1.00$$

Correct signs for z or comment or diagram

m1



Prob between 10 and 20

Correct method - requires first M mark only

M1

$$0.98778 - 0.84134 = 0.146$$

$$0.146 \text{ (} 0.146 - 0.147 \text{)}$$

A1

4

- (b) 80% within $28 \pm 1.2816 \times 8$
1.2816 or 1.282 or 1.28

Their $z \times \sigma$

B1

M1

17.7 to 38.3 mins

M1

Method using their z

A1

$$28 \pm 10.3 \text{ (} 10.2 - 10.3 \text{)}$$

$$\text{or } 17.7 \text{ (} 17.7 - 17.8 \text{)}$$

$$\text{and } 38.3 \text{ (} 38.2 - 38.3 \text{)}$$

4

[12]

Q17.

(a) $z_1 = \frac{125 - 129}{2.5} = -1.6$

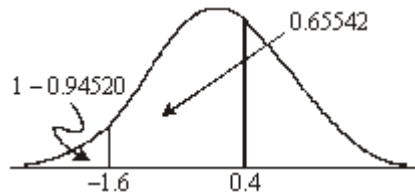
Method for z ignore sign; ignore attempt at continuing correction

M1

$$z_2 = \frac{130 - 129}{2.5} = -1$$

both z's correct sign

m1



Probability between 125 and 130 = $0.65542 - (1 - 0.94520) = 0.601$

A1

5

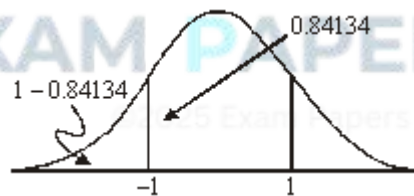
- (b) (i) 127.5
127.5 or 127 or 128 only

B1

1

$$(ii) \quad z_1 = \frac{130 - 127.5}{2.5} = 1$$

$$z_2 = \frac{125 - 127.5}{2.5} = -1$$



Their mean

M1

Probability between 125 and 130

$$= 0.84134 - (1 - 0.84134) = 0.683$$

Completely correct method (dependent on M1 and 127.5)

0.683 (0.68 – 0.685)

m1 A1

3

- (c) $125 + 2.3263 \times 2.5 = 130.8$
2.3263 (2.32 – 2.33)

their $z \times 2.5$
m1 correct method
130.8 (130.8 – 130.82) or 131

B1 M1 m1 A1

4

[12]

Q18.

(a) $z_1 = \frac{(30 - 34)}{5} = -0.8$
Method for z – ignore sign
Both signs correct

M1 m1

$z_2 = \frac{(40 - 34)}{5} = 1.2$

$P(30 < X < 40) = 0.88493 - (1 - 0.78814)$
Any correct use of normal tables - generous
Completely correct method
0.673 (0.6725 – 0.6735)

M1 m1 A1

5

- (b) 90% of journeys will exceed
1.2816 or 1.282 or 1.28

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$34 - 1.2816 \times 5 = 27.6$
Candidate's $z \times 5$
Completely correct method
27.6 (27.55 – 27.65)

M1 m1 A1

4

[9]

Q19.

(a) (i) $z_1 = \frac{(60 - 40)}{10} = 2$
Method for z (ignore sign, ignore errors in converting
9.00 am into minutes after 8.00 am)

M1

Probability before 9.00 am = 0.97725
a correct use of normal tables
 0.977(0.977 – 0.978)

M1/A1

3

$$(ii) \quad z_2 = \frac{(30 - 40)}{10} = -1$$

Probability between 8.30 am and 9.00 am
 = 0.97725 – (1 – 0.84134) = 0.819
completely correct method
 0.819(0.818 – 0.819)

m1/A1

2

(b) $35 + 2.3263 \times 2 = 39.65$
 $2.3263(2.32 \times 2.33)$
their z × 2

B1/M1

Leave at 8.20 am
method for number of minutes before 9.00 am
method for time
 8.20(8.20 – 8.21)

m1/m1/A1

5

(c) Maximum if aims for 8.45 arrival time

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E2(1)

2

i.e. leave 28 minutes before 8.45 am

8.17 am

(d) (i) Shortest mean time

E1

(ii) Least variable

E1

2

[14]

Q20.

(a) (i) $z = (185 - 65)/60 = 2.0$
method for z

M1

probability $< 185 = 1 - 0.97725 =$
completely correct method

M1

0.02275
 $0.0227 - 0.023$

A1

3

(ii) $z_1 = (50 - 65)/60 = -0.25$

$z_2 = (125 - 65)/60 = 1.0$
method for both z's including correct sign

M1

Probability between 50 and 125 minutes
any correct use of normal tables

M1

$= 0.84134 - (1 - 0.59871)$
completely correct method

M1

$= 0.440$
 $0.439 - 0.441$

A1

4

(b) Mean only just over one standard deviation above zero.

Normal distribution would give a substantial chance of negative time which is impossible.

allow both marks if point clearly made

B2

2

[9]

Q21.

(a) (i) $z = (28 - 23)/4 = 1.25$

attempt at z - ignore sign
completely correct method

M1

P(less than 28) =

M1

0.894

$0.894 \sim 0.895$

A1

3

(ii) $z_1 = (20 - 23)/4 = -0.75$
correct method, both z's

M1

$z_2 = (30 - 23)/4 = 1.75$
any correct use of normal tables

M1

P(between 20 and 30)
completely correct method

M1

$= 0.95994 - (1 - 0.77337)$

$= 0.733$

$0.733 \sim 0.734$

A1

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4

(b) $23 + 2.3263 \times 4$
2.3263 or 2.326 or 2.33 or 2.32
any correct expression, their z
completely correct method, their z

B1 M1 M1

$= 32.3$ minutes

$32.3 - 32.31$

A1

4

(c) $23 \pm 1.6449 \times 4 =$
1.6449 or 1.645 or 1.64 or 1.65

B1

$$23 \pm 6.58$$

their $z \times 4$

completely correct method, their z

M1 M1

$$16.4 \sim 29.6$$

$$6.575 \sim 6.585 ; 16.4 \sim 16.45 ;$$

$$29.55 \sim 29.6$$

A1

4

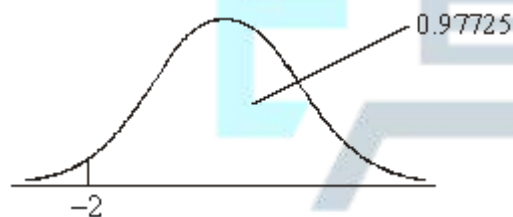
[15]

Q22.

(a) (i) $z = \frac{111 - 135}{12} = -2$

*Attempt at z , ignore sign,
disallow variance*

M1



Any correct use of normal tables

M1

$$P(> 111) = 0.977$$

$$0.977 - 0.978$$

A1

3

(ii) $z_1 = \frac{141 - 135}{12} = 0.5$

$$z_2 = \frac{150 - 135}{12} = 1.25$$

Correct method, both z s, ignore signs

M1



Correct use of normal tables
Completely correct method

m1 m1

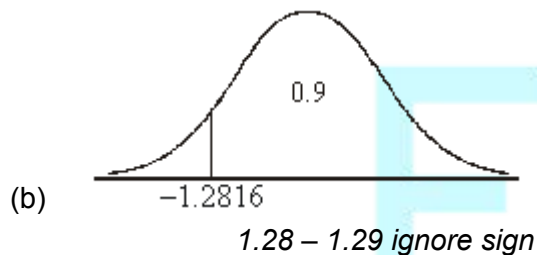
Probability between 141 and 150

$$0.89435 - 0.69146 = 0.203$$

$$0.202 - 0.204$$

A1

4



B1

Longest journey $135 - 1.2816 \times 12$

Correct method – allow incorrect sign
Completely correct method

M1 m1

= 120 miles

$$119 - 120$$

A1

4

[11]

Q23.

(a) $z_1 = \frac{250 - 266}{32} = -0.5$

Attempt z ignore sign

M1

$$z_2 = \frac{300 - 266}{32} = 1.0625$$



Any correct use of normal tables
Correct method for both z's

M1 m1

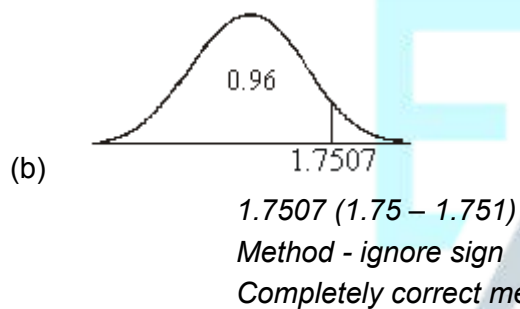
Probability between 250 and 300

$$0.8560 - (1 - 0.69146) = 0.547$$

Completely correct method
0.547 (0.546 – 0.548)

m1 A1

5



B1 M1 m1

$$k = 266 + 1.7507 \times 32 = 322$$

$$322 (321.5 - 322.5)$$

A1

4

[9]

Q24.

$$z = \frac{(902.5 - 901)}{2} = 0.75$$

Method for z - ignore sign
Any correct use of normal tables

M1 M1



m1

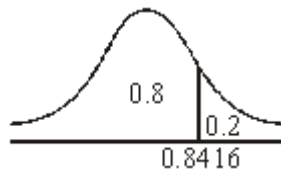
Probability $< 902.5 = 0.773$
 $0.773(0.773 - 0.774)$

A1

[4]

Q25.

(a)



$0.8416(0.841 - 0.842)$

Use of $z \times 6$.

B1 M1 m1 A1

Completely correct method

4

$25.0(25.0 - 25.1)$

(b) (i) $\frac{6}{\sqrt{90}} = 0.632$

$\frac{6}{\sqrt{90}}$ acf

B1

1

(ii) $z = \frac{(21 - 20)}{\frac{(6)}{\sqrt{90}}} = 1.581$

Use of $\frac{6}{\sqrt{90}}$.

Method for z – ignore sign

M1 m1



Probability mean exceeds 21 is

$$1 - 0.9431 = 0.0569$$

Completely correct method

$$0.0569(0.0568 - 0.0571)$$

M1 A1

4

[9]



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